

The U-Shaped Challenge of Language Grounding: How Instruction Granularity Induces VLA Shortcuts

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Abstract

Instruction granularity remains an uncontrolled variable in language-guided embodied AI, leaving its impact unknown. We propose an agent-centric formalization of granularity as rule width (w), a cross-task metric adapted from width-based planning. By measuring the peak concurrent state-feature dependencies an agent must resolve, w provides a rigorous measure of the extent to which an instruction reduces the decision-making burden required to map language into actions. We also introduce MINI-BEHAVIOR-GRAN, a controlled benchmark systematically varying w across 20 tasks while holding perception and control constant. Operationalizing granularity as rule width (w) reveals a non-monotonic, U-shaped performance curve: while fine-grained instructions ($w \leq 1$) ease grounding, coarse ones ($w \geq 2$) induce “language-ignoring” shortcuts driven by high instruction predictability ($P(l|v)$). We further uncover a generalization asymmetry, where coarse-trained models transfer to fine instructions but not vice-versa, and show that mixed-granularity training is essential for robust language grounding.

1 Introduction

Despite the rapid progress in language-guided embodied AI, the community has yet to establish a consensus on a unified metric to quantify the granularity of instructions during language grounding. In this domain, language serves a specific functional role: it provides structural information to decompose complex tasks and reduce the search space of the underlying sequential decision-making process. Consequently, *instruction granularity* should not be conflated with superficial syntactic properties, such as sentence length or the number of entities mentioned; rather, it must rigorously reflect the *decision-making burden* that remains for the agent to resolve internally after grounding the language into the action space.

Existing research conceptualizes instruction granularity through two dominant lenses, both of which struggle to provide a consistent measure of the decision-making burden required for grounding. Studies in Natural Language Processing (NLP) typically operationalize granularity via surface-form properties, such as textual specificity or discourse structure (e.g., number of entities mentioned) (Dong and Lapata, 2018; Huang et al., 2025b). While these effectively measure syntactic complexity, they adopt a *text-centric* rather than *agent-centric* view. Consequently, they cannot quantify how much an instruction reduces an embodied agent’s *decision-making burden*.

From an *agent-centric* perspective, the essential function of instruction is to decompose complex tasks into tractable sub-problems, thereby reducing the planning burden (Drexler et al., 2022). Recent work in LLM-based agents thus treats granularity as *decomposition depth*—the extent to which an instruction enumerates subgoals versus stating only the final objective (Yang et al., 2024; Myers et al., 2024; Shi et al., 2025; Zhong et al., 2025). While this aligns with our view that *instruction granularity* modulates planning demand, decomposition depth remains an unstable, task-dependent proxy. It conflates instructional guidance with the task’s inherent structure: complex tasks with interleaved subgoals may resist breakdown (e.g., the Sussman Anomaly (Sussman, 1975)), while trivial ones can be artificially elongated.

To operationalize this, we draw inspiration from the concept of policy sketches (Drexler et al., 2024) and formalize instruction granularity as rule width (w) (Bonet et al., 2019). We argue that w provides a rigorous proxy for the decision-making burden that an instruction imposes on an embodied agent. To this end, the contributions of our paper are:

(1) Width as a Formalism for Instruction Granularity: We formalize instruction granularity as *rule width* (w), adapting a concept from width-

based planning (Bonet et al., 2019; Drexler et al., 2024) that measures the peak number of concurrent state variables (features Φ) that must be tracked to fulfill the instructions. We argue w measures the *intrinsic decision complexity* of a task-instruction pair, regardless of whether the agent is symbolic or neural (§ 3).

(2) A Controlled Benchmark: We present MINI-BEHAVIOR-GRAN, the first testbed to isolate instruction granularity from confounding factors like task horizon and perception. It enables the first study of how instruction granularity affects VLA performance (§ 4).

(3) The U-Shaped Trend and Width Cap: We identify a non-monotonic U-shaped relationship where performance peaks at the Fine ($w = 0$) and Coarse ($w \geq 2$) extremes. We also identify a performance collapse at high width ($w \geq 4$): agents fail to fulfill instructions requiring maintaining concurrent features ≥ 4 , revealing a sharp language grounding boundary of existing VLA models (§ 5).

(4) Language-Ignoring Shortcuts: We diagnose the performance rebound at $w \geq 2$ as “Language Ignoring,” where agents bypass linguistic grounding to exploit visual shortcuts due to high instruction predictability ($P(l|v)$). We show that *Axial Training* (mixed-granularity) mitigates this failure mode. Additionally, we identify an *asymmetric generalization* pattern: agents trained on coarse instructions successfully transfer to fine-grained ones, but not vice versa.

2 Related Work

Instruction Abstraction and Granularity. In this work, *instruction granularity* refers to the extent to which state and action constraints are explicitly specified versus left for the agent to infer when grounding language into actions. This is distinct from, yet often conflated with the concept of *linguistic abstraction*. The term *abstraction* is viewed as *representational compression*—using a compact description to stand for multiple instances or executions. This definition is common in prior work of computational thinking (Wing, 2010; Lachmy et al., 2022; Zheng et al., 2024). Importantly, granularity can vary independently of abstraction: “tidy the room” is coarse but non-abstract (it leaves the exact required actions underspecified, yet it lacks representational compression.), whereas “repeat wiping until clean” is relatively specific yet abstract (it prescribes an action but uses a loop condition to

compress multiple executions).

While not framed as the term “granularity”, we acknowledge that a rich body of work has focused on augmenting language specificity and its impact, for example through coarse-to-fine semantic parsing (Dong and Lapata, 2018; Li et al., 2020). Other work generates compositional instructions via constraint dropout and sub-task recombination (Huang et al., 2025b), or construct counterfactual instructions to augment data diversity (Glossop et al., 2025).

In embodied instruction following, varying instruction detail is often handled via a simple two-part decomposition: high-level plans are converted into sequences of low-level steps before being grounded into actions (Yang et al., 2024; Shi et al., 2025). Most recent VLA works, however, operate with static language instructions (Ma et al., 2025; Li et al., 2025). Some models like InstructVLA and ThinkAct explores having chain-of-thought reasoning steps before action decoding (Yang et al., 2025; Huang et al., 2025a). These hierarchical definitions remain task-specific and qualitative, lacking a unified, cross-task metric to establish shared granularity levels across tasks. Consequently, quantitatively controlled studies on the formal dynamics of instruction granularity remain scarce in the embodied AI domain.

Language Sensitivity vs. Ignoring in VLA. Recent VLA research highlights a dichotomy between fragile language sensitivity (Akula et al., 2022) and systematic language-ignoring (Xing et al., 2025). We reconcile these via rule width (w): low w (Fine) involves explicit feature constraints and low predictability ($P(l|v)$), which enforces the agent to rely on language cues. Conversely, $w \geq 2$ (Coarse) triggers ignoring via visual shortcuts as $P(l|v)$ increases. Crucially, we identify a hard capacity cap at $w \geq 4$, where agents fail to track concurrent state-features regardless of granularity, revealing a fundamental grounding boundary in current architectures.

3 Rule Width as Instruction Granularity

3.1 Running Example

We first illustrate *planning width* computation using a running example. Table 1 shows the desired state progressions of a task where two shoes are cleaned using a rag, and the rag is placed on the floor afterwards. The key **environment dynamics** are: (1) the rag must be soaked before it can clean

State Progression	Concurrent Features Tracked (Sequence of Tuples)	Width
Initial State	$[\langle \rangle]$	0
Navigate to rag	$[\langle p_r = X \rangle, \dots, \langle p_r = 0 \rangle]$	1
Pick up rag	$[\langle p_r = 0 \rangle, \langle H_r \rangle]$	1
Navigate to sink	$[\langle H_r, p_s = X \rangle, \dots, \langle H_r, p_s = 0 \rangle]$	2
Turn on tap (sink)	$[\langle H_r, p_s = 0 \rangle, \langle T_s \rangle]$	2
Soak rag	$[\langle H_r, T_s, p_s = 0 \rangle, \langle S_r \rangle]$	3
Navigate to Shoe 1	$[\langle H_r, S_r, p_{f1} \downarrow \rangle, \dots]$	3
Clean Shoe 1	$[\langle H_r, S_r, p_{f1} = 0 \rangle, \langle C_1 \rangle]$	3
Navigate to Shoe 2	$[\langle H_r, S_r, C_1, p_{f2} \downarrow \rangle, \dots]$	4
Clean Shoe 2	$[\langle H_r, S_r, C_1, p_{f2} = 0 \rangle, \langle C_1, C_2 \rangle]$	4
Put rag on floor	$[\langle C_1, C_2, H_r \rangle, \dots, \langle C_1, C_2, onfloor_r \rangle]$	3

Table 1: State progression and width for the “Cleaning k Shoes” ($k = 2$). Width scales as $w = k + 2$.

shoes, and (2) the tap (sink) must be turned on to soak the rag. We represent states using a set of task-relevant feature variables (Φ): (i) boolean features H_r (holding rag), T_s (tap (sink) is turned on), S_r (rag is soaked), C_i (shoe i is cleaned), $onfloor_r$ (rag is on the floor); and (ii) spatial features p_r, p_s, p_f (distance to the rag, sink, and nearest uncleaned shoe, respectively).

First, approaching the rag is a navigation-only segment: the desired state progression is to decrease the distance feature p_r until $p_r = 0$. Thus, the width is 1 as only a single feature needs to be tracked. In this environment dynamics, $p_r = 0$ enables the state transition to H_r . Thus, the rag can be obtained at $p_r = 0$ and the width remains 1. Subsequently, H_r must be maintained while progressing to next states, resulting in a width of 2. The rag becoming soaked requires simultaneous maintenance of three features, i.e., at the sink ($p_s = 0$), holding the rag (H_r), and having the tap (sink) turned on (T_s), which yields a width of 3. After soaking, C_1 can be obtained by tracking H_r , S_r , and p_f , giving width 3. The subtlety arises when cleaning the second shoe: one has to track the same features as before, plus maintaining the cleanliness of the first shoe (C_1), which increases the width to 4.

Why no action model? Agents’ action sets constrain how state transitions are executed, but *width* focuses on state-level feature dependencies induced by the environment dynamics. Different agents (e.g., a rover vs. a drone) may have different action spaces to achieve the same feature change, so emphasizing state progressions promotes generalization of width-based concepts across different embodiments.

The impact of *instruction granularity* on VLA behavior remains underexplored largely because the field lacks a **formal, quantifiable, cross-task**

measure of granularity. Building on the limitations of heuristic metrics discussed in § 1, we formally equate *instruction granularity* with *latent planning demand*, i.e., the implicit decision-making burden required to ground text into executable actions. We operationalize this using *rule width* (w) by framing natural-language instructions as policy sketches (Drexler et al., 2024). Under this framework, fine-grained instructions decompose tasks into minimal-width sub-problems, whereas coarse instructions leave a large “grounding gap” that the agent must resolve internally.

3.2 Symbolic Task Representation

While VLAs operate on raw multimodal observations, we adopt a symbolic, relational state abstraction. Many embodied benchmarks provide such structured representations—object properties and relations—that can be expressed as *fluents* to form symbolic state or goal conditions (Shridhar et al., 2020; Li et al., 2024). Formally, we consider a classical planning problem $P = \langle \mathcal{F}, \mathcal{A}, I, G \rangle$, where $\mathcal{F}$ is a set of fluents, $\mathcal{A}$ a set of actions, I the initial state, and G the goal.

To bridge raw states and high-level instructions, we define a set of **features** $\Phi = \{\phi_1, \dots, \phi_n\}$, where ϕ_i is a Boolean or integer-valued function over state space S . These features serve as domain-level abstractions that capture task-relevant aspects of the environment. Thus, each s is associated with a **feature valuation vector** $\phi(s) = (\phi_1(s), \dots, \phi_n(s))$.

3.3 Instructions as Rule Sets $\mathcal{R}$

To study instruction granularity in a controlled manner, we formalize each instruction a symbolic interpretation as a set of “if-then” rules $\mathcal{R} = \{r_1, \dots, r_m\}$ defined over the feature space Φ . This rule-based representation does not aim to capture the full expressiveness of natural language; instead, it distills the constraint/goal structure that governs the computational cost incurred by a symbolic search algorithm (i.e., planning demand). Specifically, each r_i is of the form $C_{r_i} \mapsto E_{r_i}$:

- C_{r_i} (the *condition*) is a conjunction of feature constraints. For a Boolean feature p , a constraint is p or $\neg p$; for a numerical feature n , it is $n = 0$ or $n > 0$.
- E_{r_i} (the *effect*) is a conjunction of feature changes. For a Boolean feature p , an effect is

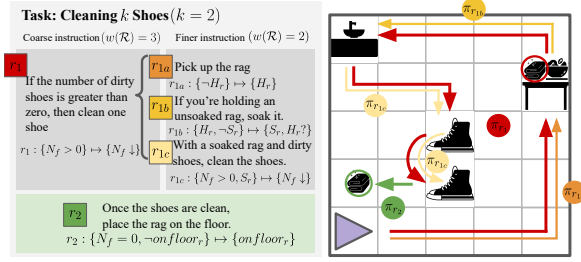


Figure 1: (Left) Factoring r_1 into r_{1a} - r_{1c} reduces latent planning complexity via explicit action guidance. (Right) Fine instructions segment the long-horizon coarse trajectory (π_{r_1} , red).

p , $\neg p$, or $p?$ (allowing any change); for a numerical feature n , it is $n \downarrow$ (decrease), $n \uparrow$ (increase), or $n?$ (any change).

These forms capture the core semantics of the instructions we study. A rule r_i is **fulfilled** by a state sequence $(s, \dots, s')$ if $\phi(s) \models C_i$ and the pair $(\phi(s), \phi(s'))$ satisfies the changes in E_i (e.g., $\forall(n \downarrow) \in E_{r_i}, \phi_n(s') < \phi_n(s)$) while keeping unmentioned features constant (i.e., $\phi_j(s') = \phi_j(s)$ for $j \notin E_i$). To ensure that the agent’s performance strictly reflects its grounding ability rather than the unsolvability of the task, we guarantee that all instruction rule sets $\mathcal{R}$ are executable and dead-end-free. Formally, this is achieved by ensuring the rules are well-formed (i.e., they terminate at the goal) and satisfy the soundness and closure properties (Bonet and Geffner, 2021).

3.4 Rule Width as a Granularity Measure

The width $w(r_i)$ of a rule r_i represents the number of state variables (features) that must be tracked concurrently to fulfill the transition. Formally, for a rule to have width k , the agent must coordinate at most k features from Φ to reach the effect E_i from a state satisfying C_i (Lipovetzky and Geffner, 2012).

We define the granularity of an instruction $\mathcal{R}$ by its width-based planning bottleneck: the maximum width among its constituent rules: $w(\mathcal{R}) = \max_{r_i \in \mathcal{R}} w(r_i)$.

We continue the running example to demonstrate two instruction sets, each representing a different width level, with two variants visualized in Figure 1. We also append a **counting feature** in Φ to track the number of uncleaned shoes, denoted as N_f .

Coarse-Grained Instruction ($w(\mathcal{R}) = 3$) This instruction states only the high-level rules and their

desired ordering, leaving the internal structure of each sub-task implicit:

- $r_1 : \{N_f > 0\} \mapsto \{N_f \downarrow\}$ (clean a shoe; $w = 3$)
- $r_2 : \{N_f = 0, \neg on_floor_r\} \mapsto \{on_floor_r\}$ (put the rag on the floor after cleaning shoes; $w = 1$)

Middle-Grained Instruction ($w(\mathcal{R}) = 2$) This variant decomposes r_1 to explicitly state the *soaking* process:

- $r_{1a} : \{\neg H_r\} \mapsto \{H_r\}$ (pick up the rag; $w = 1$)
- $r_{1b} : \{H_r, \neg S_r\} \mapsto \{S_r, H_r\}$ (soak the rag)
- $r_{1c} : \{N_f > 0, S_r\} \mapsto \{N_f \downarrow\}$ (clean; $w = 2$)
- $r_2 : \{N_f = 0, \neg on_floor_r\} \mapsto \{on_floor_r\}$ (put the rag on the floor; $w = 1$)

This rule set is well-formed: it is goal-terminating via r_2 , and the rules are sequentially reachable, meaning the successor state of any fulfilled rule (that is not the goal) also meets the condition of at least one rule in $\mathcal{R}$. The set is also robust to perturbations: if an agent drops the rag after soaking it ($\neg H_r$), the rule set remains applicable by reverting to r_{1a} to reacquire the rag before proceeding to the cleaning task (r_{1c}).

Width as a Planning Proxy. The fundamental purpose of decomposing a task into a rule set $\mathcal{R}$ is to provide structural scaffolding, guiding the agent through intermediate “stepping stone” states. By design, we assume the execution of r_i does not negate established fluents unless explicitly required by E_i . This is the essence of language-guided decomposition: the agent treats features in C_i as *maintained*, allowing working memory to focus on *incremental* features. Without this assumption, language guidance would offer no reduction in complexity; the agent would need to track every feature since the initial state.

This intuition is formally grounded in novelty-driven Breath First Search (Lipovetzky and Geffner, 2017). By pruning states that fail to achieve a new feature conjunction of size $\leq k$, search complexity is bounded by $O(|\Phi|^k)$. Thus, trajectories that unnecessarily negate an already-achieved precondition C_i (unless required by E_i) are longer than those preserving it, and are pruned from the optimal frontier. Consequently, features in C_i that are already true and not altered by E_i are effectively maintained, and the active coordination w reduces to the novel features needed to reach the

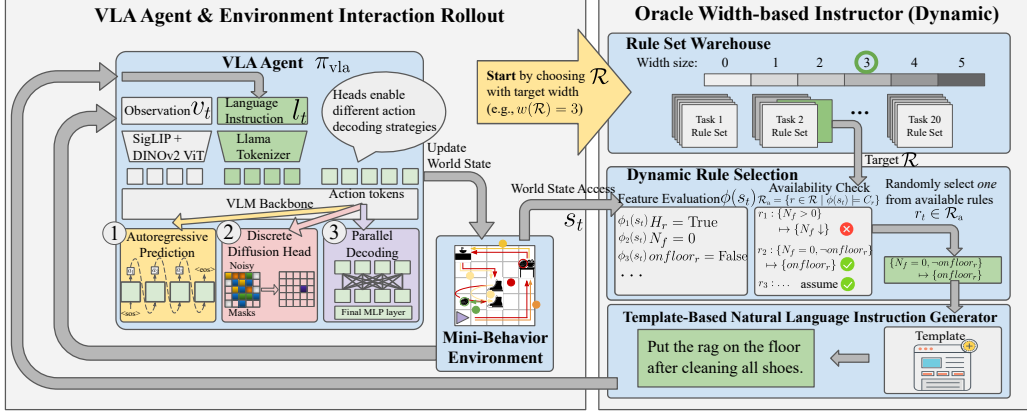


Figure 2: MINI-BEHAVIOR-GRAN Experimental Pipeline. We evaluate the impact of instruction granularity across three VLA action-decoding strategies. During inference time, an Oracle Instructor evaluates the current world state to decide available rule set $\mathcal{R}_a$ and active rule r_t . A rule-based natural language instruction l_t is generated accordingly. The process iterates until completion or max steps.

next subgoal. For instance, in r_{1b} (soaking), H_r is established by r_{1a} ; dropping it prematurely would merely lengthen the path to S_r , so active coordination is restricted to $\{p_s = 0, T_s\}$, yielding $w = 2$.

To communicate these rules to VLA agents, we translate each rule into a natural-language sentence via a template-based generator. These templates verbalize the features within each rule; e.g., $\{H_r, \neg S_r\} \mapsto \{S_r\}$ is realized as “If you are holding the rag but the rag is not yet soaked, soak the rag.” This construction preserves the granularity differences defined by the rule width. Additional language prompt examples are provided in § F.

4 Mini-BEHAVIOR-Gran Benchmark

To disentangle the influence of instruction granularity in VLA performance, we seek an environment that (1) retains long-horizon and heterogeneous activities so that instruction guidance meaningfully modulates the planning demand, while (2) simplifies perception and low-level control to avoid confounding factors. As such, we use MINI-BEHAVIOR (Jin et al., 2023) as our testbed. Detailed justification for our choice over other environments is provided in § B.

We introduce MINI-BEHAVIOR-GRAN, the first VLA benchmark specifically designed to enable a controlled study and formal quantification of instruction granularity via *rule width*. Specifically, we define a shared, domain-level feature set Φ that captures task-relevant attributes across all 20 complex household tasks on top of the symbolic state abstraction from the original bench-

mark. For each task, we design rule sets with varying widths (up to six levels). We verify that all rule sets are well-formed and solvable. We utilize an Oracle Instructor to deliver instructions during inference. This ensures that any observed performance drops are strictly attributable to the model’s inability to ground specific granularity levels, rather than stochastic variations in instruction quality or applicability that would confound our cross-granularity analysis (see Figure 2 right panel). At each time t , it evaluates the current feature vector $\phi(s_t)$, identifies the set of available rules $\mathcal{R}_a = \{r \in \mathcal{R} \mid \phi(s_t) \models C_r\}$. We sample an active rule $r_t \in \mathcal{R}_a$ and it is then processed by a template-based translator to generate a natural language instruction l_t , to match the VLA model’s input format. l_t is paired with the current observation v_t and fed into the VLA model to predict action chunk $(a_t, \dots, a_{t+k}) \sim \pi_{\text{vla}}(\cdot \mid v_t, l_t)$.

5 Experiments

5.1 Experimental Setup

Task Suite & Training. We use MINI-BEHAVIOR-GRAN, which contains 20 complex tasks sharing the same feature set Φ . We further classify instruction granularity into three *relative* classes of *Fine* (F), *Medium* (M), and *Coarse* (C) via per-task quantile binning. In most cases, this mapping corresponds to $w = 0$ for Fine, $w = 1$ for Medium, and $w \geq 2$ for Coarse. We evaluate seven training settings: (1) **Pure F/M/C** (1 : 0 : 0): trains exclusively on one granularity class; (2) **Centroid** (1 : 1 : 1): uniformly sample from all three classes; (3) **Axial F/M/C** (3 : 1 : 1):

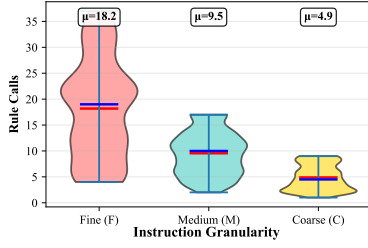


Figure 3: Rule call distribution across instruction granularities. # of rule calls decreases with coarser instructions. It decouples instruction length from width.

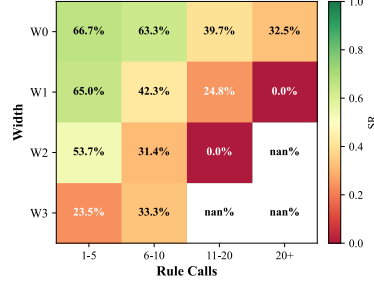


Figure 4: Impact of width and horizon on SR. SR decays with task horizon; larger instruction width imposes an additional burden even when horizon is fixed.

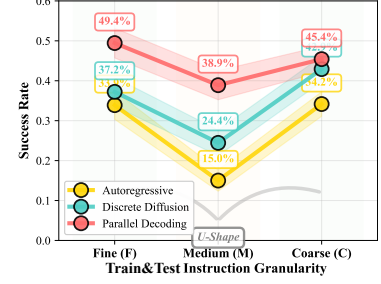


Figure 5: In-Distribution (matched train/test) SR across granularities (i.e., train Fine $\rightarrow$ test Fine, etc.). A non-monotonic U-shaped trend appears.

emphasizes one class with some exposure to the rest classes. Our dataset provides 50 training and 10 evaluation instances per task, each paired with reference trajectories generated by BFWS width-based planner from Lipovetzky and Geffner (2017) (see § E for details).

Action-Decoding Archetypes. The primary architectural difference among latest VLA models lies in their action-decoding strategies (Zhong et al., 2025). Hence, we employ a unified VLM backbone but vary three representative action-decoding strategies (see Figure 2 left panel): (1) **Autoregressive** (AR), exemplified by RT-2 (Zitkovich et al., 2023), OpenVLA (Kim et al., 2024) and others (Reed et al., 2022; Huang et al., 2024; O’Neill et al., 2024) that predict actions sequentially via causal attention; (2) **(Discrete) Diffusion**, which uses diffusion process for action chunk prediction (Black et al., 2025; Bjorck et al., 2025; Shukor et al., 2025). We employ discrete diffusion (Liang et al., 2025) for compatibility with the discrete action space of our testbed; (3) **Parallel Decoding**, utilizing bidirectional attention for single-pass action prediction to improve throughput, as demonstrated by OpenVLA-OFT (Kim et al., 2025b), PD-VLA (Song et al., 2025) and other variants (Yin et al., 2025; Wang et al., 2025; Xiao et al., 2025). Details on model hyperparameters, hardwares and training schedules are in § G.

5.2 Establishing the Yardstick: Decoupling Horizon from Width

A critical confounder in evaluating VLA planning complexity is the task sequence length. However, in language-guided execution, instructions inherently decompose long-horizon tasks into segments. Consequently, the relevant measure of sequential

burden shifts from the raw atomic horizon (total low-level action steps) to the instruction horizon, i.e., the number of rule calls issued by the instructor.

To strictly isolate rule width (w) from sequential burden, we train all agents on the same set of tasks. This ensures that the distribution of atomic horizons (total low-level steps) encountered during training is held constant across agents, leaving the number of rule calls and instruction granularity as variables: fine-grained instructions ($w = 0$) aggressively decompose the task into numerous low-width rule calls, resulting in a low average steps per rule call; coarse instructions ($w \geq 2$), by contrast, yield fewer rule calls with a higher average steps per call (Figure 3).

To validate w as a robust proxy for latent planning demand for learning-based agents, we analyze Figure 4 using the *Centroid* configuration, which ensures balanced exposure across all granularity levels. The heatmap reveals a strict monotonic decay in Success Rate (SR) as rule width increases, confirming that larger rule widths fundamentally amplify the grounding complexity of the instruction-to-action mapping.

5.3 The U-Shaped Paradox and Capacity Cap

To establish a baseline, we train models using the *Pure F/M/C* setting. the models are trained and evaluated on the same granularity class. As shown in Figure 5, all VLA archetypes exhibit a non-monotonic *U-shaped performance curve*.

At the fine extreme ($w = 0$), where instructions require minimal latent reasoning to be grounded into actions, models achieve high performance. Performance then decreases at medium granularity ($w = 1$), before unexpectedly rebounding at the

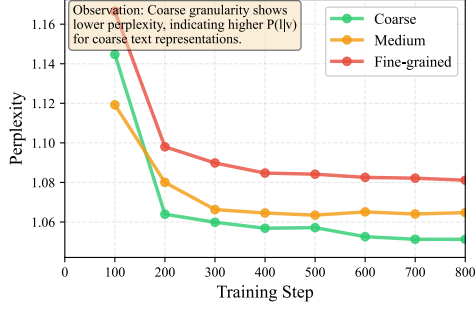


Figure 6: Perplexity of a VLM model trained to predict instructions from visual context, compared across granularities. Coarse instructions are more predictable, supporting high $P(l | v)$ claim.

coarse regime ($w \geq 2$). This contradicts the assumption that higher planning width should monotonically degrade learning.

Crucially, beyond this U-shaped regime, we identify a hard capacity cap at $w \geq 4$ (shown in Figure 9). For tasks requiring the concurrent tracking of four or more state features, all models fail regardless of the action head archetypes or training settings. This exposes a fundamental language grounding bottleneck in current VLA models, demonstrating that w precisely marks this capacity boundary.

5.4 Mechanistic Probe: The Grounding Valley

Why does performance rebound at $w \geq 2$ while collapsing at $w = 1$? We demonstrate this is driven by a competition between *instruction predictability* and *latent planning complexity*.

Predictability and Perplexity. As granularity coarsens, text diversity decreases. Consider a surjective mapping f from a fine-grained label space L_{fine} to a coarser one L_{coarse} . The conditional probability of a coarse label aggregates its constituent fine labels: $P(c_j | v) = \sum_{l_i \in f^{-1}(c_j)} P(l_i | v)$.

Consequently, coarse-grained instructions are statistically more predictable from visual contexts alone. We empirically validate this by fine-tuning a Vision-Language Model (Qwen3-VL-4B via LoRA) to predict instructions from current visual observations. The resulting prediction perplexities $P(l | v)$ (1.081, 1.065, and 1.048 for Fine, Medium, and Coarse granularities, respectively) confirm a monotonic decrease as instructions become coarser, highlighting that higher-level task abstractions are better entailed by the visual state.

Language Ignoring Shortcuts. High predictability ($P(l | v)$) incentivizes multimodal models to

bypass expensive language processing (Xu et al., 2025). We quantify this “language ignoring” via $\Delta\text{SR} = \text{SR}_{w/\text{lang}} - \text{SR}_{w/o \text{ lang}}$. Models trained purely on coarse instructions exhibit a smaller gap ($\Delta\text{SR} = 20.0\%$) than fine-trained ones, confirming they develop policy invariance to text and operate as vision-dominant policies. However, attention analysis (Figure 8) reveals there is no obvious reduction in language-token attention. This suggests that in VLA architectures, maintained attention weights do not necessarily equate to language grounding, highlighting a complex dynamic that requires future investigation.

The Grounding Valley. We thus see a non-monotonic relationship between instruction predictability and language grounding complexity:

- **Fine** ($w = 0$): *Successful Grounding.* Low $P(l|v)$ (high perplexity, Figure 6) means language provides unique information. Combined with low-width grounding burden (§ 5.2), the model reliably maps instructions to actions, yielding high SR (Figure 5).
- **Medium** ($w = 1$): *The Alignment Trap.* As $P(l|v)$ rises, language becomes partially redundant, and the model reduces its reliance on text (ΔSR drops). However, unlike the coarse setting where language can be safely ignored, here language remains just useful enough to prevent the model from fully committing to a vision-only policy. The agent falls into a representational gap: language is hard to ground, yet still too present to abandon.
- **Coarse** ($2 \leq w \leq 3$): *Modality Collapse.* High $P(l|v)$ (Figure 6) causes the model to exploit visual shortcuts: ΔSR approaches 20% (Figure 7), which is a signature of modality collapse.
- **Capacity Cap** ($w \geq 4$): *Cognitive Overload.* The visual shortcuts exploited in the coarse regime fail to scale. When a task demands tracking ≥ 4 concurrent state features purely from visual (w/o explicit memory module), the planning complexity reaches $O(|\Phi|^4)$ (Lipovetzky and Geffner, 2017). Current VLA architectures cannot sustain this grounding load: performance uniformly collapses across all models at $w \geq 4$ (Figure 9), explaining the hard empirical ceiling identified earlier.

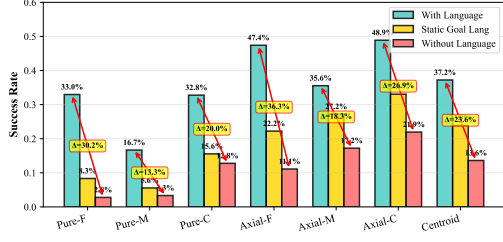


Figure 7: Both mixed and coarse-only training “generalize” better in novel instruction settings (smaller SR gap).

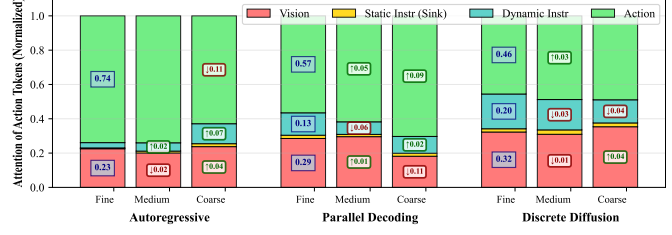


Figure 8: Attention from action tokens. Self-attention dominates, followed by vision; No clear trend across granularity levels.

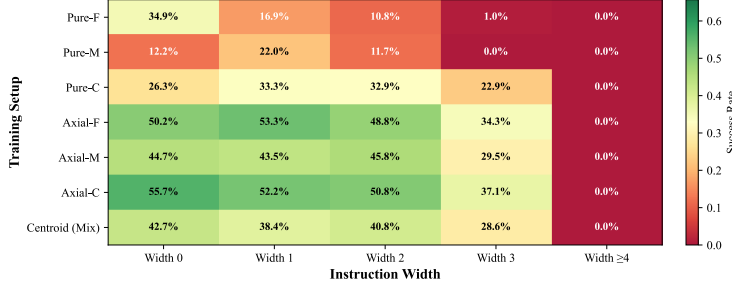


Figure 9: The x-axis are test instruction width, and the y-axis are training setup. Transfer is asymmetric: train coarse $\rightarrow$ test fine works, but train fine $\rightarrow$ test coarse does not. Mixed-granularity training improves generalization. *Axial-C* training setting performs best overall.

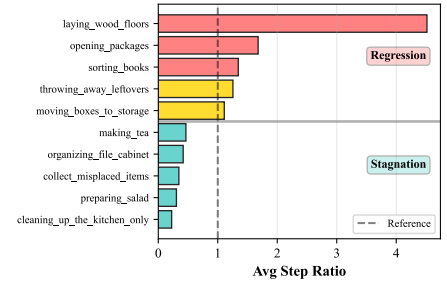


Figure 10: Low-width but long task horizon (e.g., laying woods) tends to cause regression, while high-width tasks tend to stagnate.

Mitigating Modality Collapse via Axial Training. Having diagnosed language-ignoring as a byproduct of high predictability, we hypothesize that increasing the diversity of the training language space L should artificially lower the average $P(l | v)$, thereby forcing the agent to re-engage with the text. Our mixed-granularity training results confirm this mechanism. All mixed-granularity models exhibit substantially larger ΔSR gaps compared to their *Pure-X* counterparts, indicating a successful reduction in language ignoring. Among these, the fine-weighted mixture *Axial-F* emerges as the optimal structural compromise: it achieves an overall success rate comparable to the coarse-heavy *Axial-C*, while maintaining a significantly larger ΔSR .

5.5 Other Results

Asymmetric Generalization. We observe a strict directional asymmetry (Figure 9): coarse-trained models successfully zero-shot transfer to fine texts, but fine-trained models fail on coarse texts. This corroborates our earlier ΔSR analysis: coarse training induces a vision-biased policy that generalizes robustly across instruction granularities. Conversely, fine training induces an over-reliance on dense instruction guidance, making the policy brittle when such guidance is withdrawn.

Failure Modes. Agent failure distributions bifurcate strictly with width (Figure 10). Low-width, long-horizon tasks (e.g., laying wood floors) predominantly suffer from *regression*: the agent violates previously satisfied fluents and cycles back to prior states, reflecting subgoal serialization failures. High-width tasks ($w \geq 3$), by contrast, exhibit *stagnation*: the agent freezes, unable to resolve concurrent feature dependencies ($O(|\Phi|^w)$) required to initiate a valid state transition. This dichotomy mirrors the width-based complexity landscape (see § H for details).

6 Conclusion

This work establishes rule width (w) as a rigorous, cross-task formalism for instruction granularity that quantifies the state-tracking burden of language guidance, serving as an effective proxy for language grounding difficulty in neural agents. We observed a non-monotonic relationship between granularity and performance, driven by a trade-off between instruction predictability ($P(l | v)$) and latent planning complexity ($O(|\Phi|^w)$). We demonstrate that mixed-granularity training mitigates modality collapse and preserves language grounding without sacrificing the robust visual reasoning skills learned from coarse trajectories.

7 Limitations

While this work systematically isolates the impact of language granularity on VLA planning, several boundaries constrain its current scope:

Discrete Action Spaces. We evaluate our framework in environments with discrete action spaces, a common abstraction in language-conditioned embodied AI (e.g., ALFWorld-based agent (Xiong et al., 2025; Kim et al., 2025a)) and LLM-based agent systems (Qin et al., 2024; Patil et al., 2024), where high-level reasoning is decoupled from low-level execution via skill primitives or API calls. This choice is intentional: it isolates symbolic planning from continuous control noise, allowing us to cleanly attribute observed effects to instruction granularity. However, extending this granularity-aware framework to hybrid discrete-continuous or continuous action spaces remains essential for real-world robotics applications, where low-level motor commands interact with high-level linguistic abstractions.

Predefined State Features (Φ). Our formulation of rule width relies on a predefined set of general environmental predicates. While this set captures commonsense knowledge for household activities, it assumes task-relevant state abstractions are given a priori. Future work should explore developing learnable, hierarchical feature modules that dynamically construct task-relevant state abstractions would allow the model to adjust its own “internal granularity”.

Static Visual Granularity. As a controlled study of instruction granularity, we hold visual granularity constant across conditions. Yet visual perception itself inherently operates at multiple levels of abstraction from raw pixels to object segmentation images to scene graphs. Investigating the cross-modal interplay between visual representation granularity and instruction width is a critical next step.

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